

Week 5 Practical: RBAC, NSG & Storage Security

Role-Based Access Control | Network Security Groups | Azure Storage

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Azure Tenant / Subscription:	
Date Submitted:	

Task 2 — Network Security Group (3 marks)

The screenshot shows the Microsoft Azure portal interface. On the left is the navigation pane with options like 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main area displays the 'Add inbound security rule' configuration for the 'TechBridge-NSG' Network Security Group. The configuration fields are as follows:

- Source:** Any
- Source port ranges:** *
- Destination:** Any
- Service:** Custom
- Destination port ranges:** 8080
- Protocol:** Any (selected)
- Action:** Deny (selected)
- Priority:** 4096
- Name:** Deny-All-Inbound
- Description:** (empty text box)

A warning message at the bottom states: "This rule denies traffic from AzureLoadBalancer and may affect virtual machine connectivity. To allow access, add an inbound rule with higher priority to allow...". At the bottom right, there are 'Add' and 'Cancel' buttons, and a 'Give feedback' link.

Task 3 — Storage Account & Containers (2 marks)

Microsoft Azure

Up Search resources, services, and docs (G+/)

giggles_quiver1k@idou...
DEFAULT DIRECTORY (GIGGLESQ...

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Help + support

Home > Storage center | Blob Storage

Create a storage account

BasicsAdvancedNetworkingData protectionSecurityEncryptionTags

Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below.

[Learn more about Azure storage accounts](#)

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *
Resource group *

Instance details

Storage account name * ⓘ
Region * ⓘ
Preferred storage type
Performance * ⓘ
Redundancy * ⓘ

Azure subscription 1
TechBridge-RG
[Create new](#)

techbridgestorage01
(Africa) South Africa North
[Deploy to an Azure Extended Zone](#)

Choose preferred storage type
Standard: Recommended for most scenarios (general-purpose v2 account)
Premium: Recommended for scenarios that require low latency.

Locally-redundant storage (LRS)

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Microsoft Azure

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Home > Storage center | Blob Storage

Create a storage account

BasicsAdvancedNetworkingData protectionSecurityEncryptionTags

Security

Configure security settings that impact your storage account.

Require secure transfer for REST API operations

Allow enabling anonymous access on individual containers

Enable storage account key access

Default to Microsoft Entra authorization in the Azure portal

Minimum TLS version

Permitted scope for copy operations

Microsoft Defender for Storage

Microsoft Defender for Storage is an Azure-native layer of security intelligence that detects potential threats of your storage accounts. It helps prevent the three major impacts on your data and workload: malicious file uploads, sensitive data exfiltration and data corruption.

Enable Defender for Storage

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Microsoft Azure

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Home > Storage center | Blob Storage

Create a storage account

BasicsAdvancedNetworkingData protectionSecurityEncryptionReview + create

[View automation template](#)

Basics

Subscription	Azure subscription 1
Resource group	TechBridge-RG
Location	South Africa North
Storage account name	techbridgestorage01
Primary service	
Performance	Standard
Replication	Locally-redundant storage (LRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Managed Identity for SMB	Disabled

Azure Files

Require Encryption in Transit for SMB	Enabled
---------------------------------------	---------

Security

Secure transfer	Enabled
Blob anonymous access	Disabled
Allow storage account key access	Enabled
Default to Microsoft Entra authorization in the Azure portal	Disabled
Minimum TLS version	Version 1.2

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Microsoft Azure

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Home > Storage center | Blob Storage

Create a storage account

Secure transfer	Enabled
Blob anonymous access	Disabled
Allow storage account key access	Enabled
Default to Microsoft Entra authorization in the Azure portal	Disabled
Minimum TLS version	Version 1.2
Permitted scope for copy operations	From any storage account
Enable Defender for Storage	Disabled

Networking

Public network access	Enabled
Public network access scope	Enabled from all networks
Default routing tier	Microsoft network routing

Data protection

Point-in-time restore	Disabled
Blob soft delete	Enabled
Blob retainment period in days	7
Container soft delete	Enabled
Container retainment period in days	7
Classic file share soft delete	Enabled
Classic file share retainment period in days	7
Versioning	Disabled
Blob change feed	Disabled
Version-level immutability support	Disabled

Encryption

Encryption type	Microsoft-managed keys (MMK)
Enable support for customer-managed keys	Blobs and files only
Enable infrastructure encryption	Disabled

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Create

Give feedback

Microsoft Azure | Search resources, services, and docs (G+)

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DEFAULT DIRECTORY (GIGGLESQ...

Home > Storage center | Blob Storage > techbridgestorage01

techbridgestorage01 | Containers

Storage account

Search

+ Add container

Search container

Showing all 1 items

☐	Name
☐	logs

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

- Classic file shares
- Queues
- Tables
- Security + networking
- Data management
- Settings
- Monitoring
- Automation
- Help

New container

Name *

projects ✓

Anonymous access level ⓘ

Private (no anonymous access) ▾

i The access level is set to private because anonymous access is disabled on this storage account.

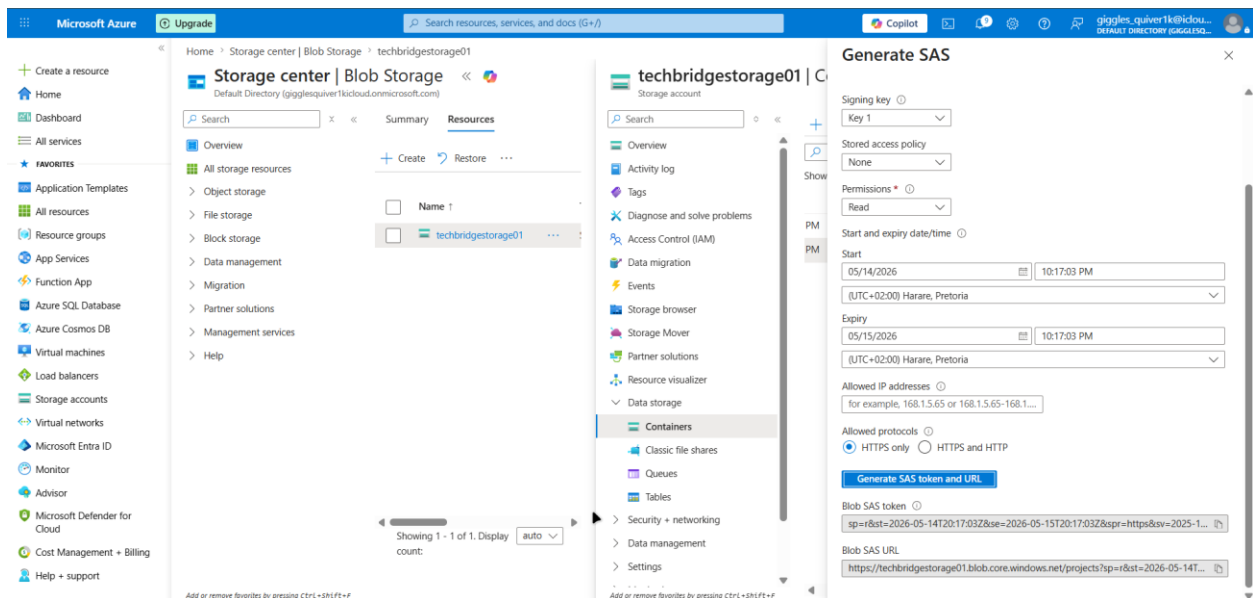
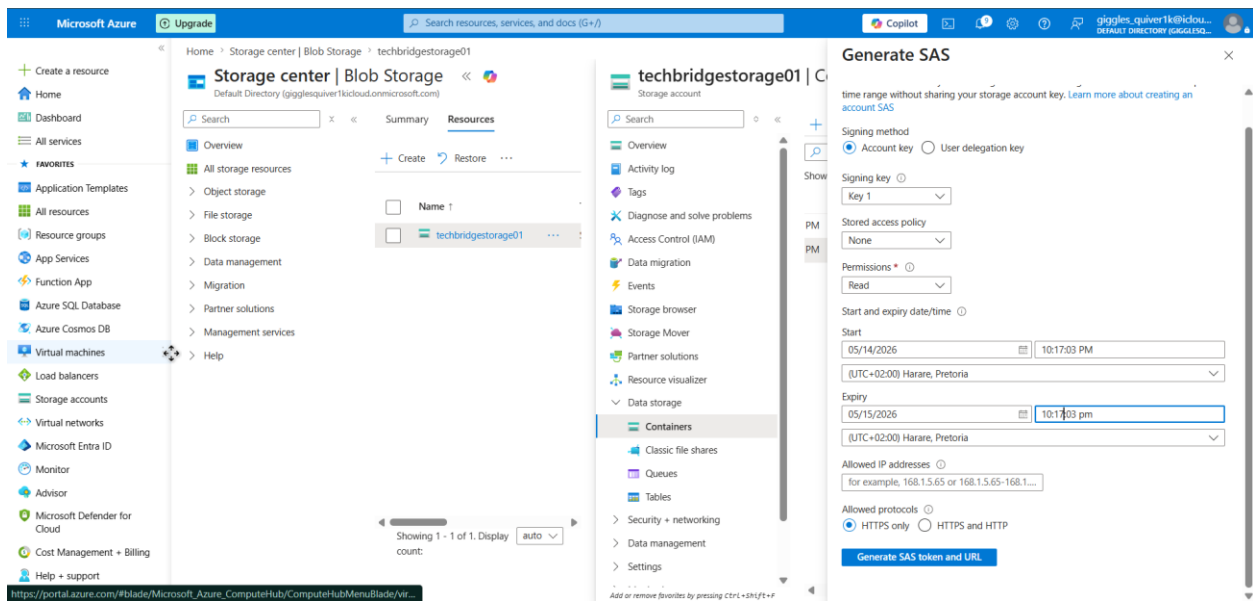
Advanced ▾

Create

Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

Task 4 — Storage Access Permissions (2 marks)



sp=r&st=2026-05-14T20:17:03Z&se=2026-05-15T20:17:03Z&spr=https&sv=2025-11-05&sr=c&sig=efuMGmplSo1Lrn%2BhLM1kKmrqm562BG9tOYYXrJACsZ8%3D

https://techbridgestorage01.blob.core.windows.net/projects?sp=r&st=2026-05-14T20:17:03Z&se=2026-05-15T20:17:03Z&spr=https&sv=2025-11-05&sr=c&sig=efuMGmplSo1Lrn%2BhLM1kKmrqm562BG9tOYYXrJACsZ8%3D